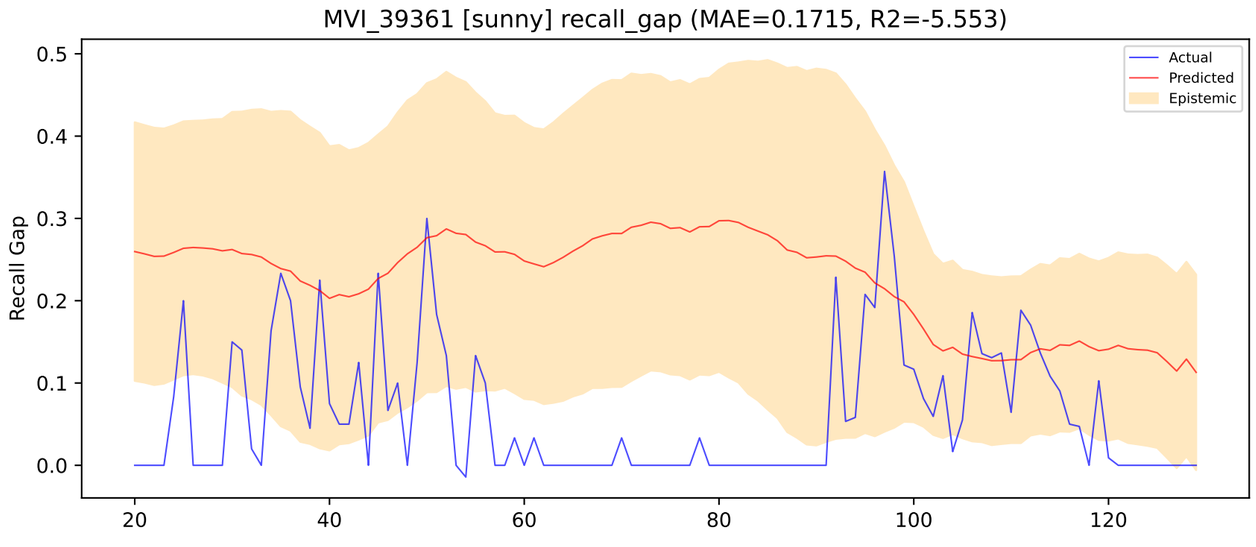
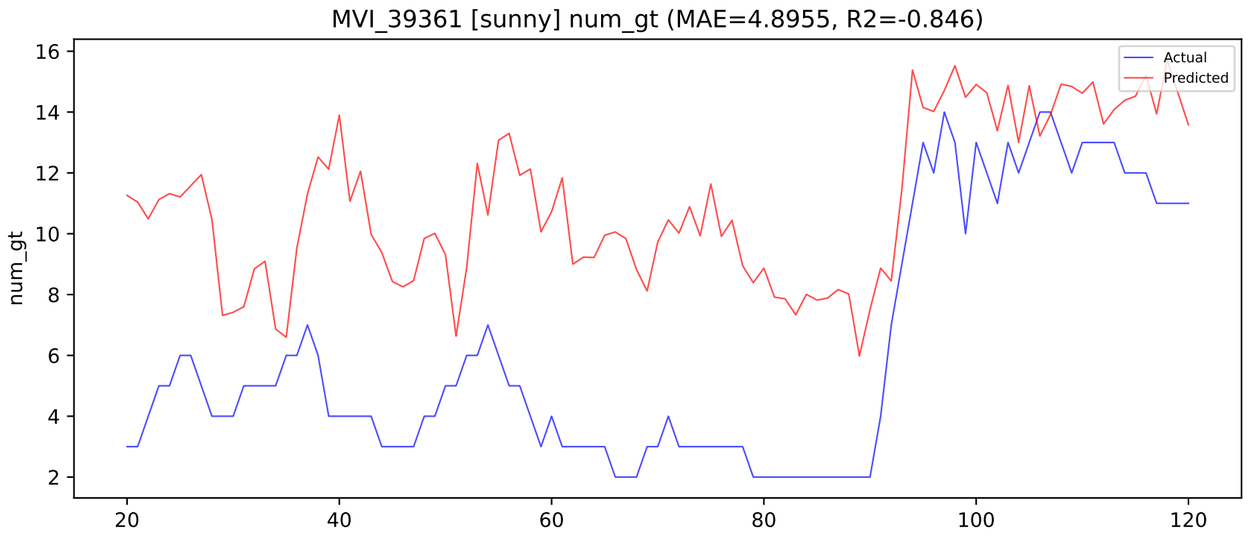
LSTM_nano (23 feat: 14 image + 9 nano detection) -- Top 2 Sequences per Weather

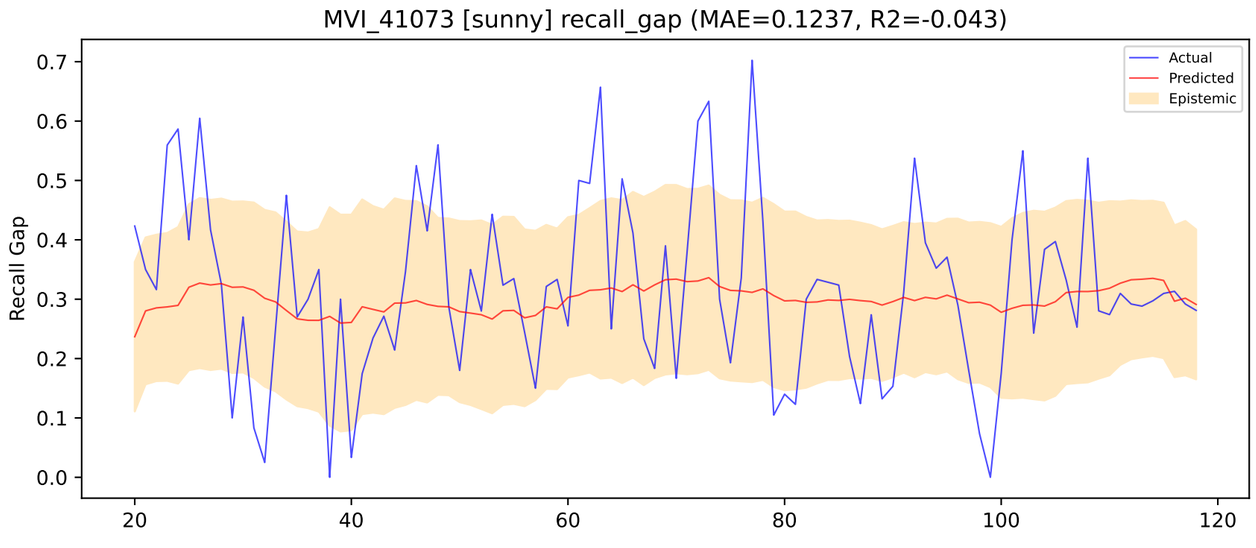
MVI_39361 [sunny] recall_gap (MAE=0.1715, R2=-5.553)



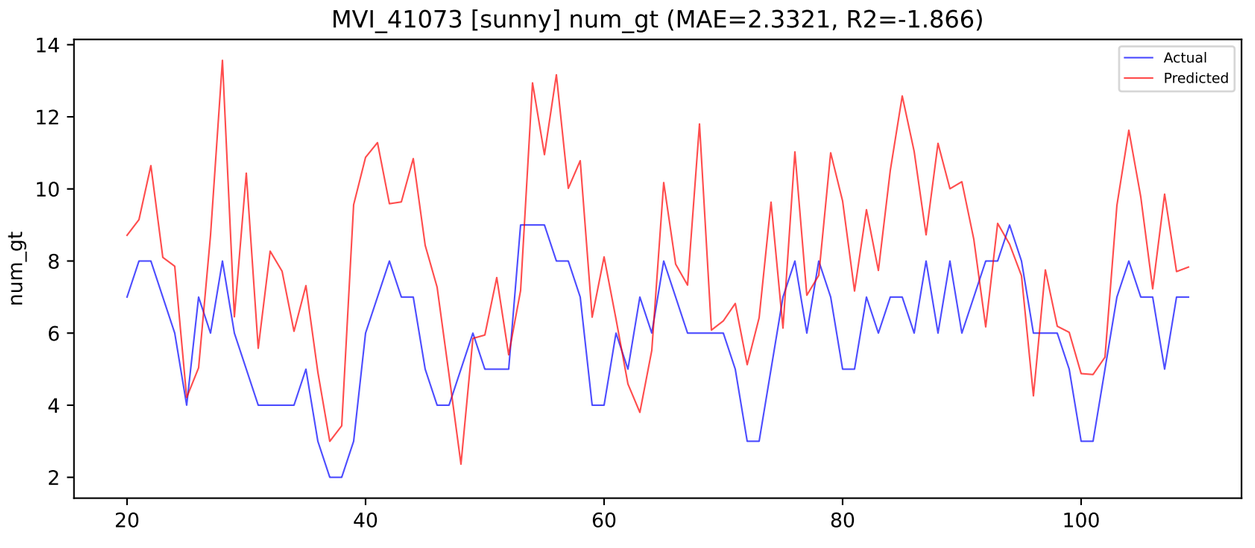
MVI_39361 [sunny] num_gt (MAE=4.8955, R2=-0.846)



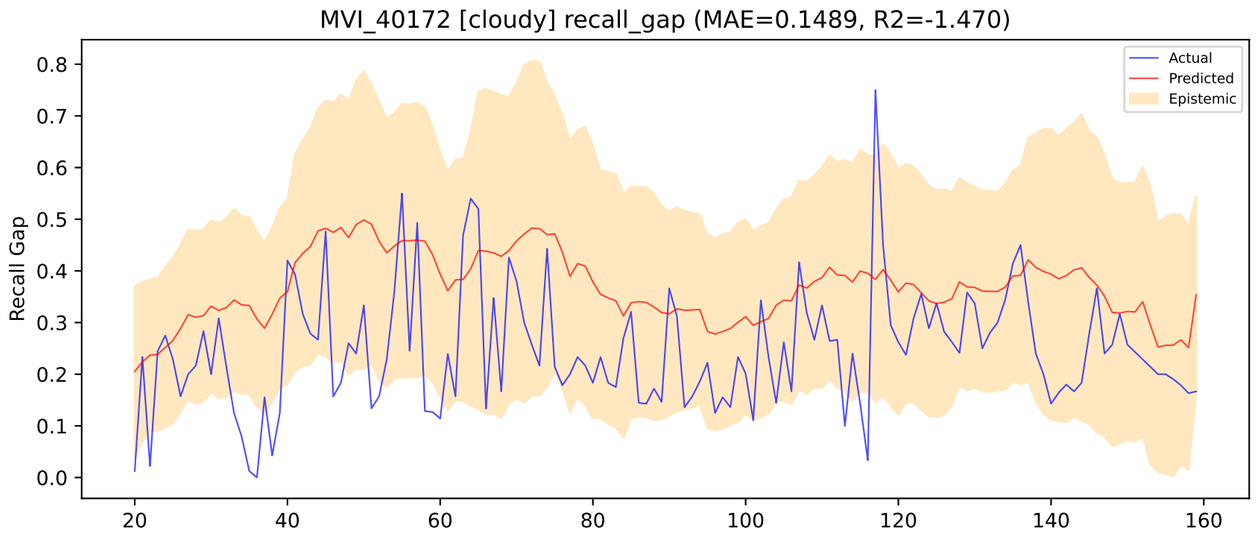
MVI_41073 [sunny] recall_gap (MAE=0.1237, R2=-0.043)



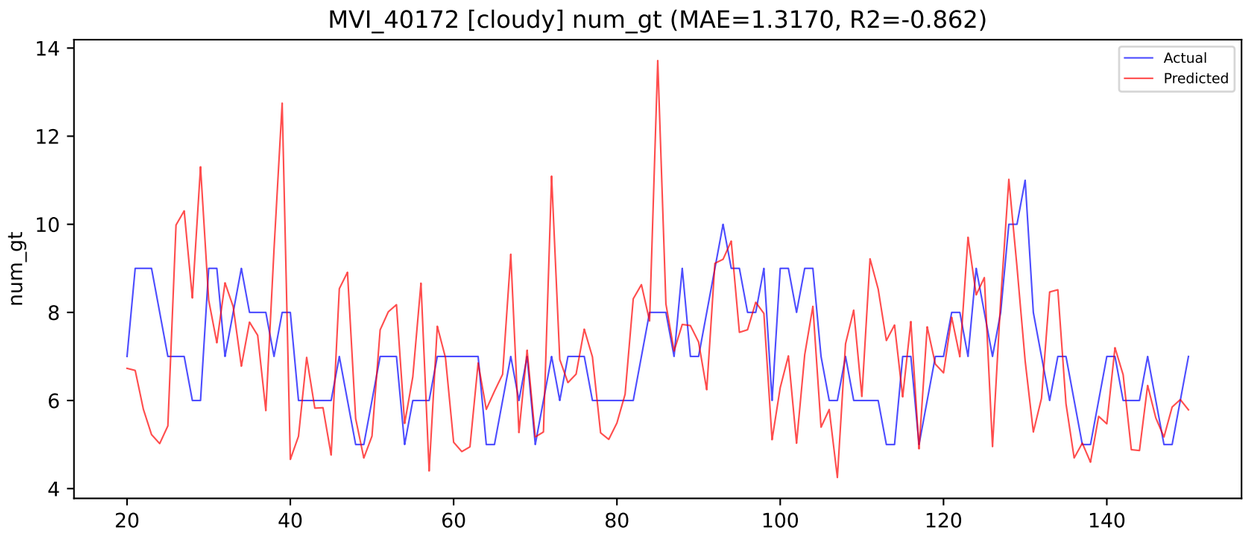
MVI_41073 [sunny] num_gt (MAE=2.3321, R2=-1.866)



MVI_40172 [cloudy] recall_gap (MAE=0.1489, R2=-1.470)



MVI_40172 [cloudy] num_gt (MAE=1.3170, R2=-0.862)



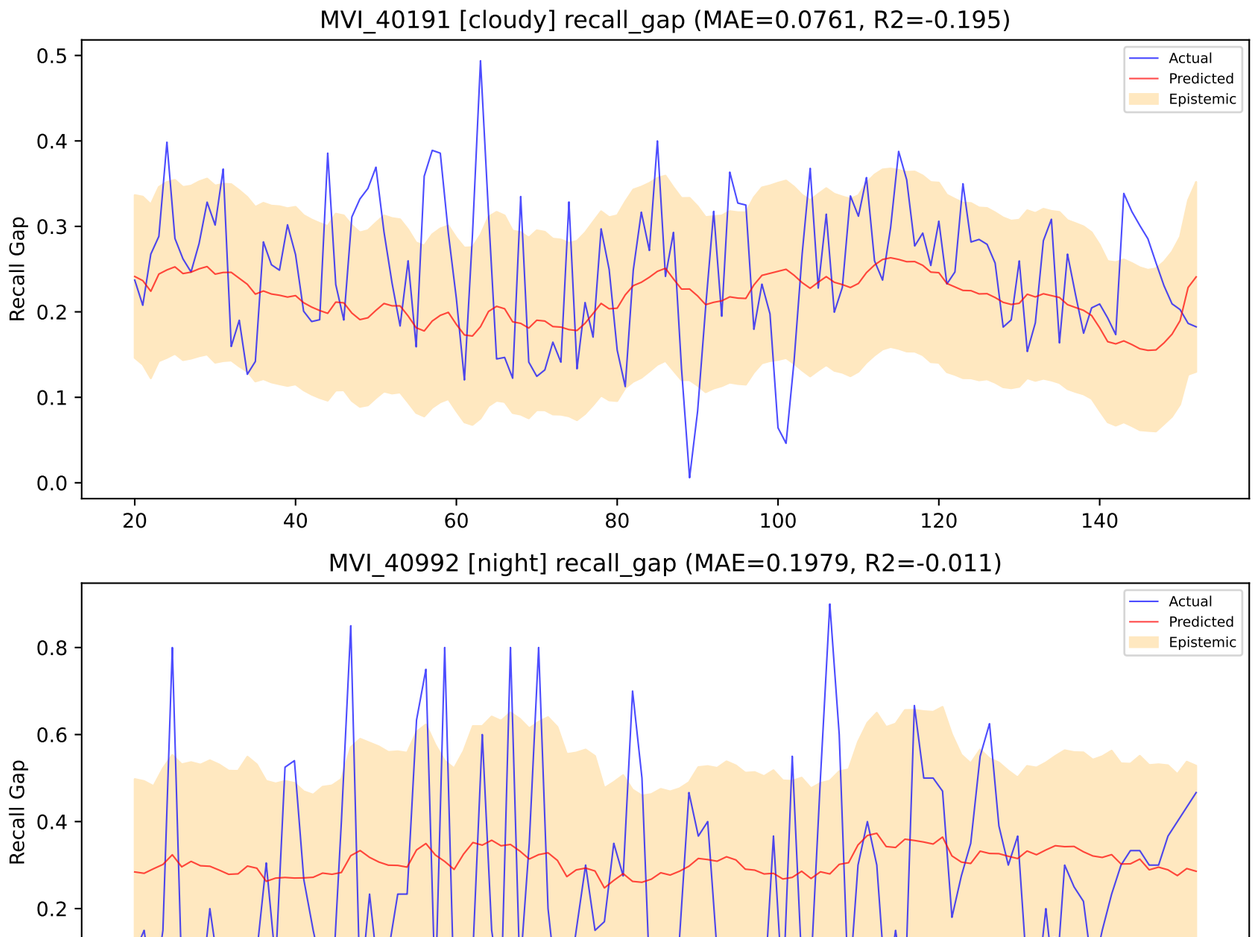
MVI_40191 [cloudy] recall_gap (MAE=0.0761, R2=-0.195)

MVI_40191 [cloudy] num_gt (MAE=2.9199, R2=-0.283)

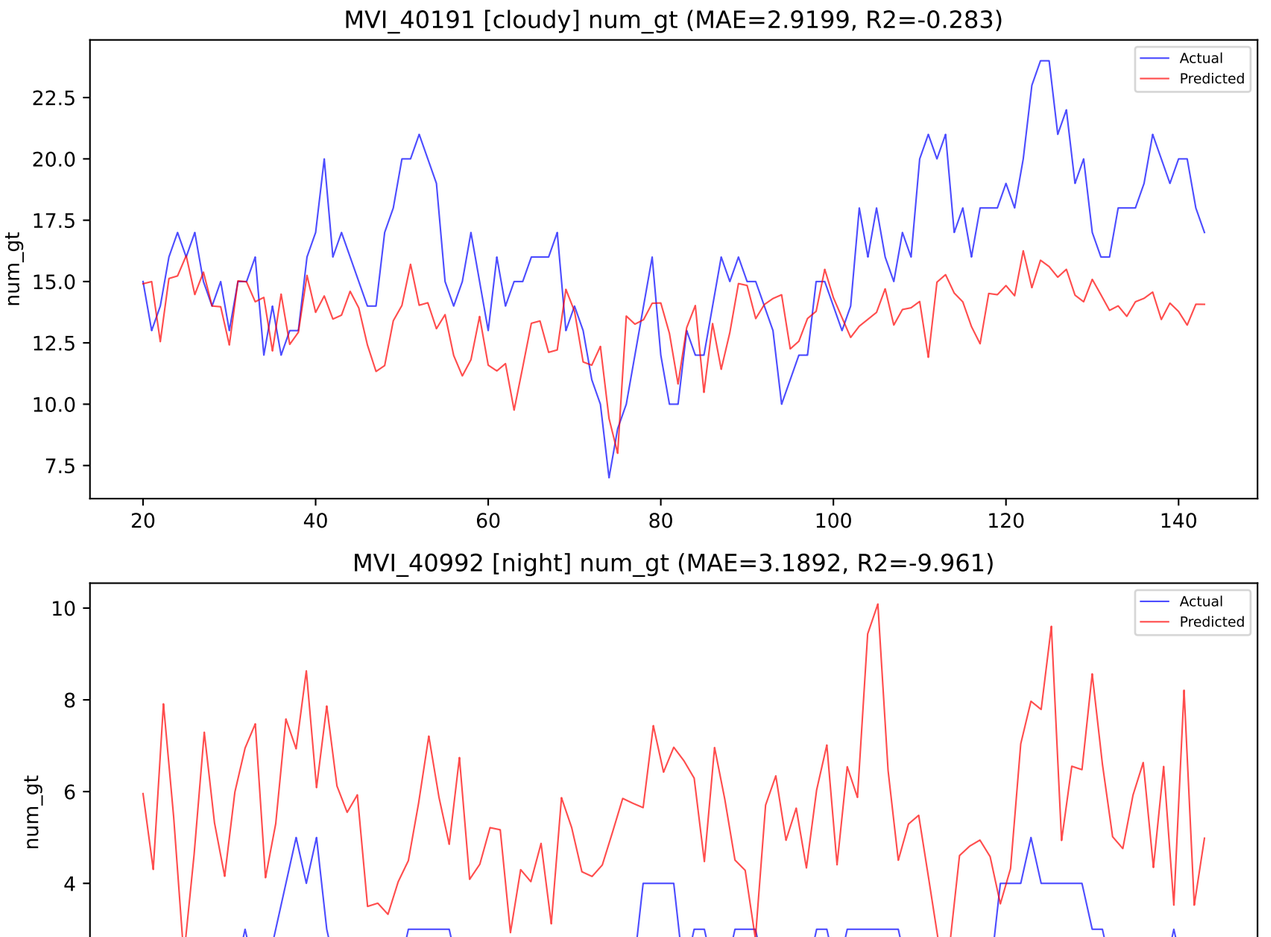
MVI_40992 [night] recall_gap (MAE=0.1979, R2=-0.011)

MVI_40992 [night] num_gt (MAE=3.1892, R2=-9.961)

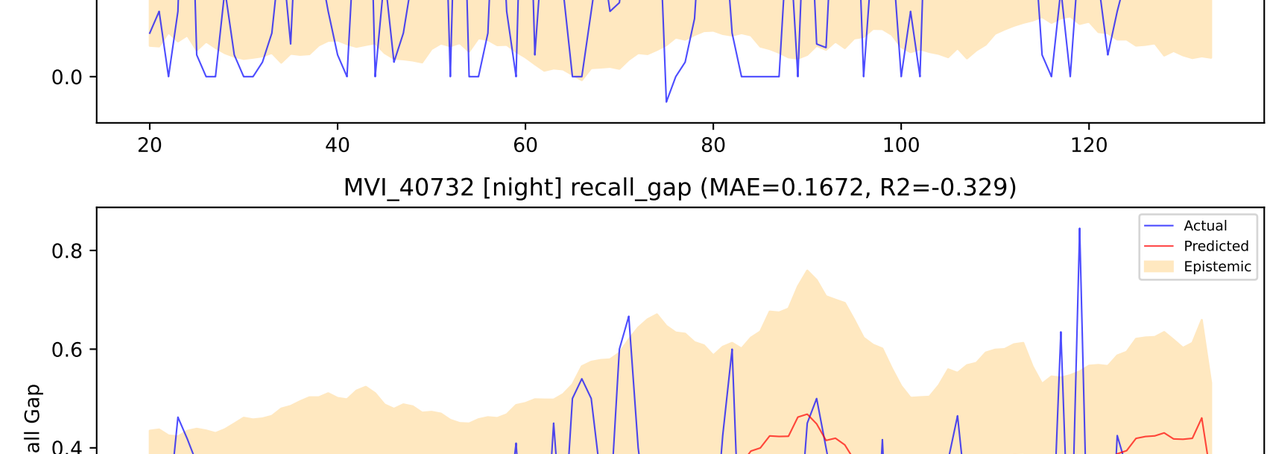
MVI_40732 [night] recall_gap (MAE=0.1672, R2=-0.329)



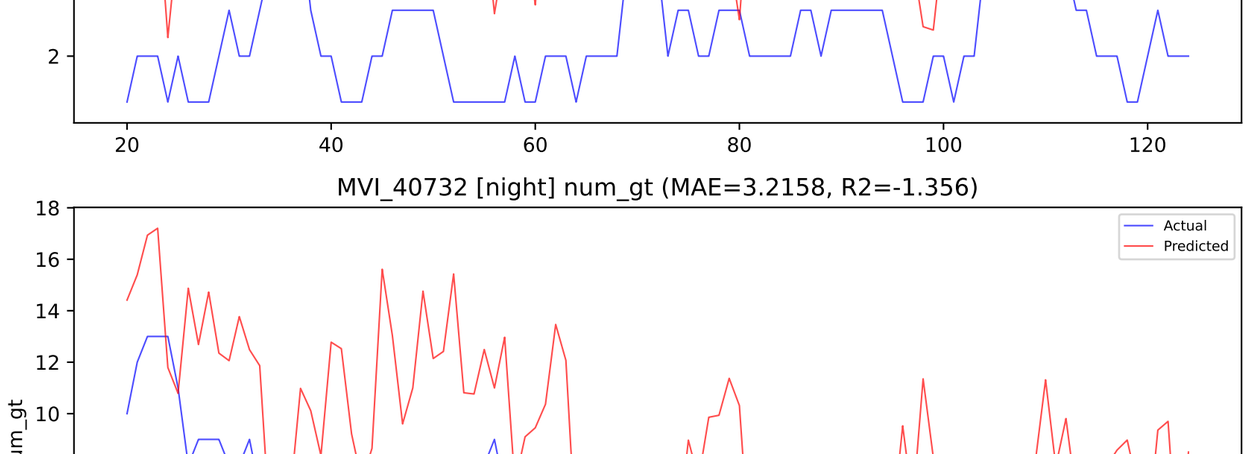
MVI_40732 [night] num_gt (MAE=3.2158, R2=-1.356)



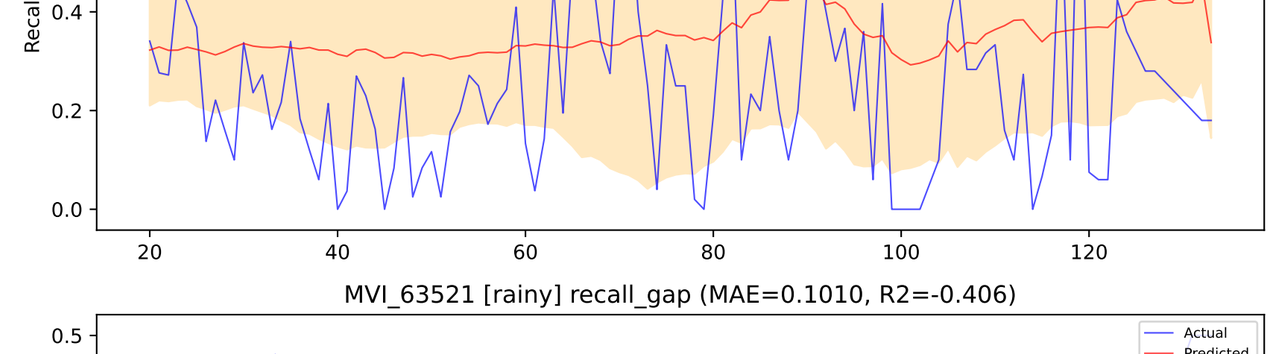
MVI_63521 [rainy] recall_gap (MAE=0.1010, R2=-0.406)



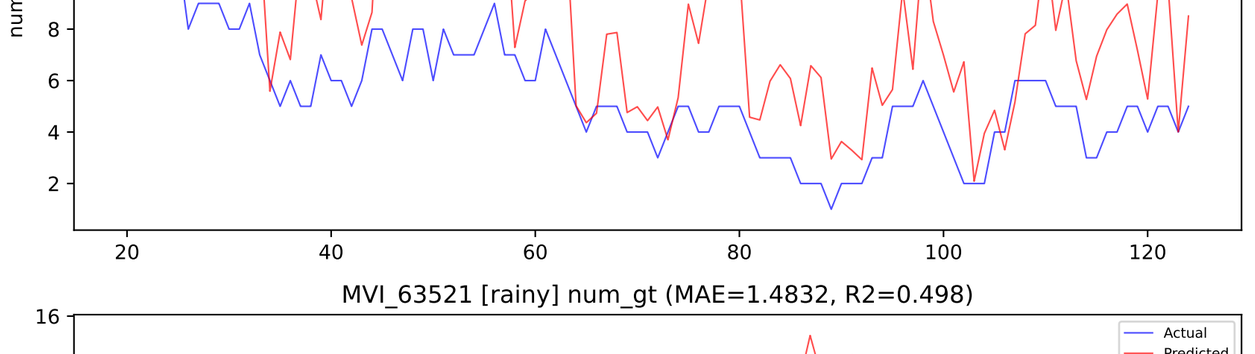
MVI_63521 [rainy] num_gt (MAE=1.4832, R2=0.498)



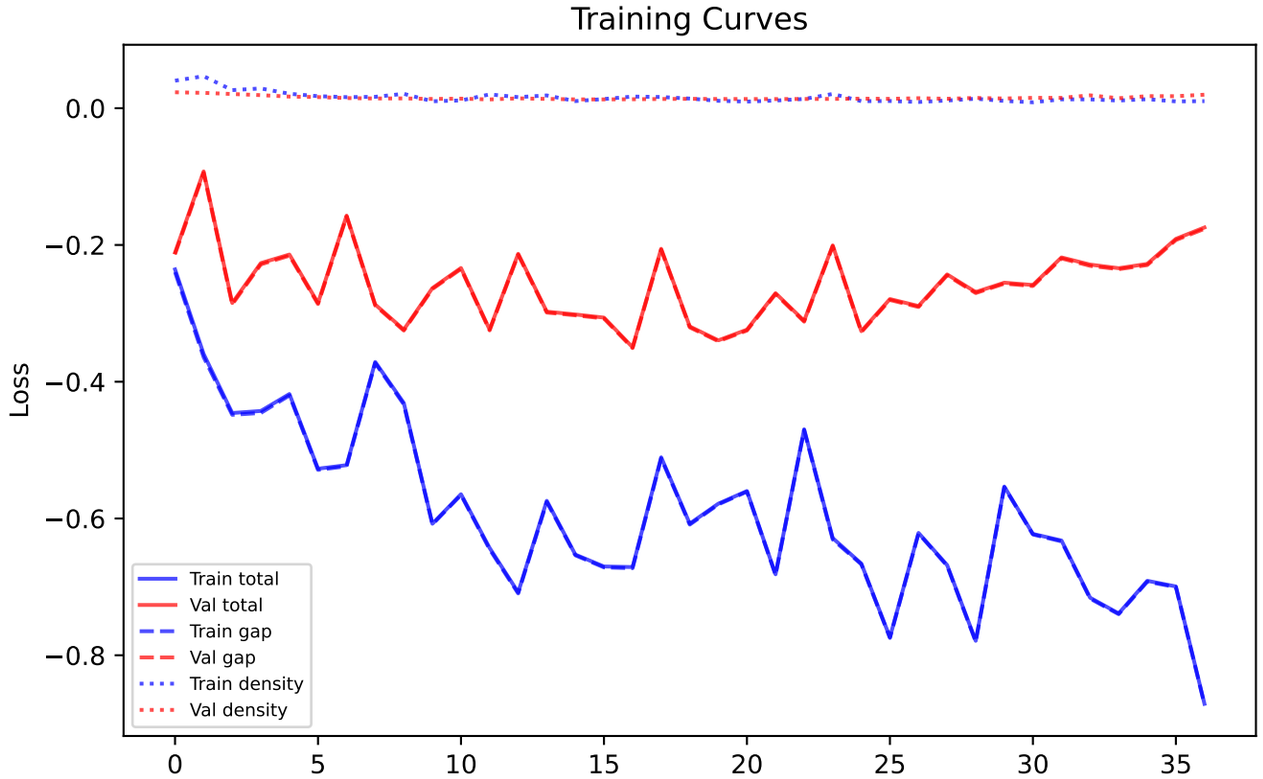
MVI_40892 [rainy] recall_gap (MAE=0.1162, R2=-1.276)

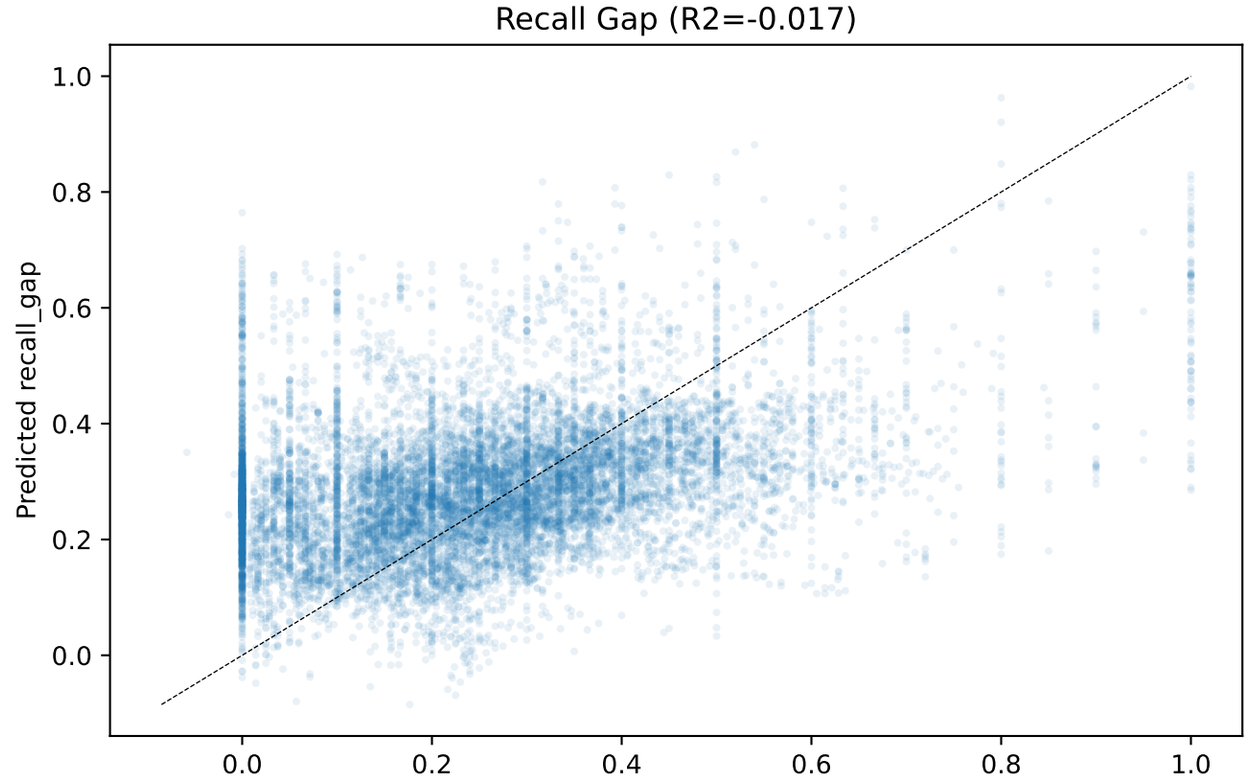


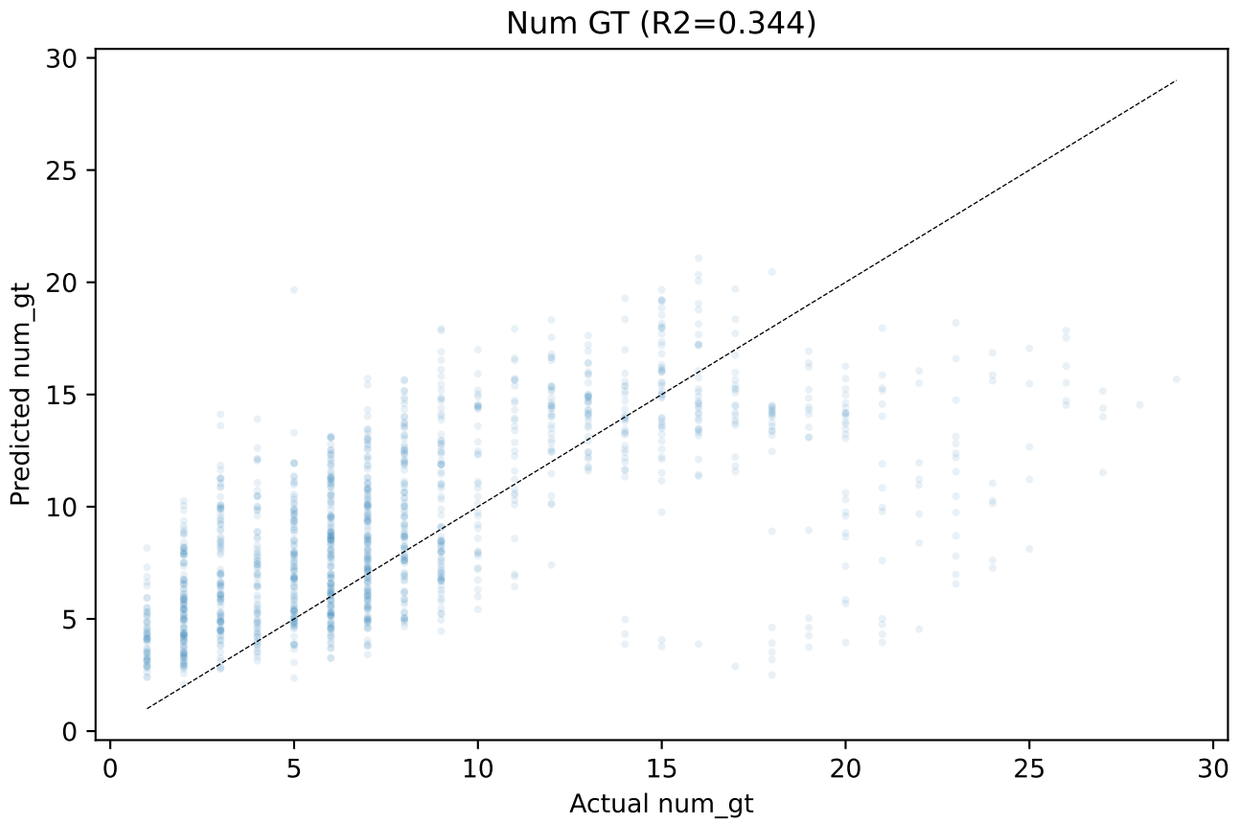
MVI_40892 [rainy] num_gt (MAE=11.2719, R2=-6.169)

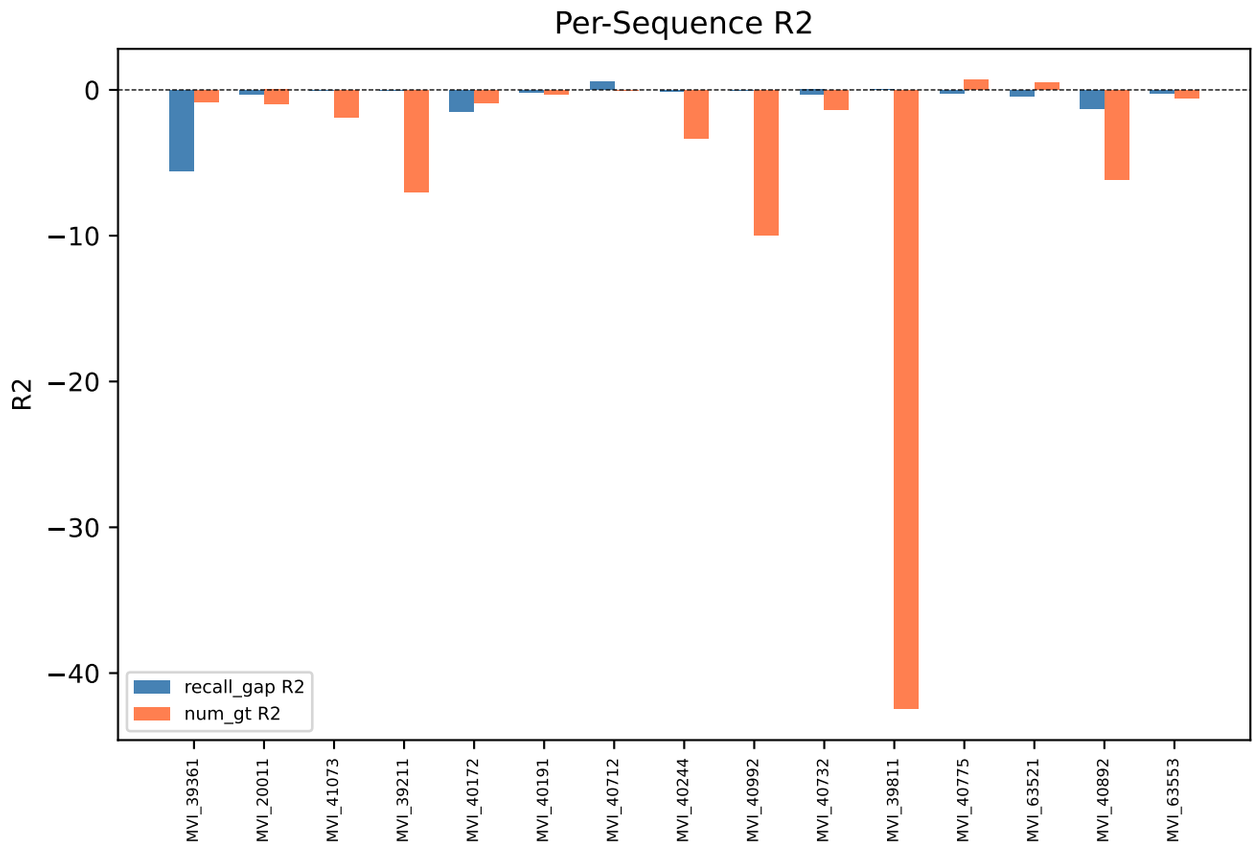


Page 2: Diagnostics









Page 3: Metrics & Comparison

LSTM_nano -- Validation Metrics (23 features: 14 image + 9 nano)

Sequence	Weather	Gap MAE	Gap R2	Gap rho	GT MAE	GT R2	GT rho
MVI_39361	sunny	0.1715	-5.5527	-0.4176	4.8955	-0.8460	0.7553
MVI_20011	sunny	0.1162	-0.2698	-0.1246	3.3657	-0.9876	0.5749
MVI_41073	sunny	0.1237	-0.0428	0.1008	2.3321	-1.8657	0.5541
MVI_39211	sunny	0.2330	-0.0443	0.4019	2.8958	-6.9999	0.4153
MVI_40172	cloudy	0.1489	-1.4703	0.3104	1.3170	-0.8623	0.3927
MVI_40191	cloudy	0.0761	-0.1946	0.3180	2.9199	-0.2832	0.4309
MVI_40712	cloudy	0.1019	0.5783	0.7601	4.0521	-0.0566	0.8494
MVI_40244	cloudy	0.1056	-0.1146	0.2654	4.1545	-3.3038	0.6129
MVI_40992	night	0.1979	-0.0105	0.2451	3.1892	-9.9609	0.4575
MVI_40732	night	0.1672	-0.3292	0.3309	3.2158	-1.3562	0.8271
MVI_39811	night	0.3007	0.0250	0.4193	2.6607	-42.4408	-0.0417
MVI_40775	night	0.0863	-0.2013	0.1725	3.1222	0.6583	0.8891
MVI_63521	rainy	0.1010	-0.4060	-0.0003	1.4832	0.4978	0.8269
MVI_40892	rainy	0.1162	-1.2758	-0.2640	11.2719	-6.1690	0.5440
MVI_63553	rainy	0.0935	-0.2489	0.1328	1.9404	-0.5648	0.6935
OVERALL		0.1365	-0.0165	0.3574	3.5106	0.3442	0.6581
BASELINE	(mean)	0.1375	0.0000	nan	4.5528	0.0000	nan

Per-Weather + Cross-Experiment Comparison

Model / Weather	Gap R2	Gap MAE	n (windows)
sunny	-0.0101	0.16704	3000
cloudy	-0.0683	0.10918	4400
night	-0.0579	0.17622	2820
rainy	0.0802	0.10422	2600
OVERALL (23 feat, nano)	-0.0165	0.13646	12820
Both models (32 feat)	0.1650	---	---
Image-only (14 feat)	-0.1150	---	---
UAVDT js10 (24 feat)	-0.2700	---	---